



VR Anatomy Learning: Kidney Visualization

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1. INTRODUCTION

❗ PROBLEM

- ❌ Traditional anatomy learning relies on static 2D diagrams.
- ❌ Students struggle to visualize 3D structures.
- ❌ Limited interaction reduces retention.
- ❌ Cadaver-based learning is costly.

💡 SOLUTION

- ✅ VR environment for interactive 3D exploration.
- ✅ Simulates organ functions and visualizes related diseases
- ✅ Enhanced retention of spatial structures.
- ✅ Supports SDG 4 (Targets 4.3 and 4.A) by improving access to quality, interactive higher education.

3. KEY FEATURES

- ✅ Immersive VR Environment
- ✅ Interactive 3D Models
- ✅ Real-time Exploration
- ✅ Disease Visualization
- ✅ Educational Experience
- ✅ User-friendly Interface

4. APPLICATIONS

- ✅ Medical Anatomy Learning
- ✅ Virtual Biology Labs
- ✅ Healthcare Training
- ✅ Disease Visualization
- ✅ Remote VR Education
- ✅ Interactive 3D Learning

6. CONTACT & SDGs

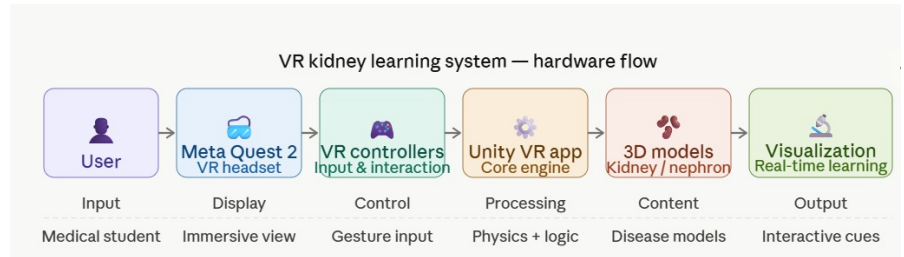


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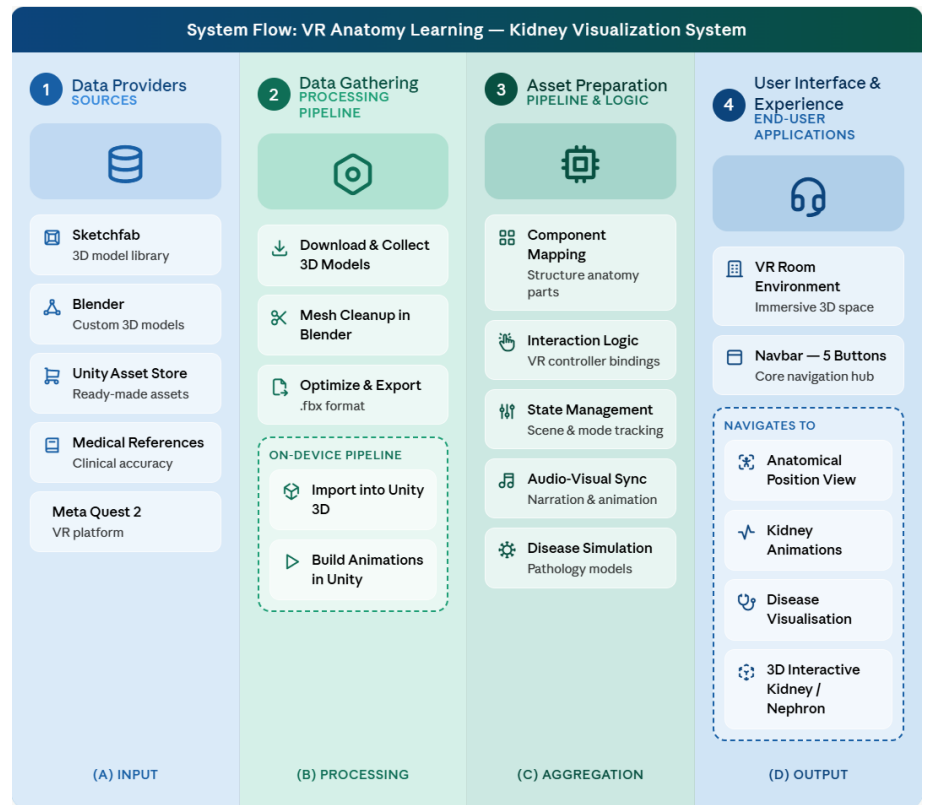


2. SYSTEM OVERVIEW

SYSTEM FLOW



SYSTEM ARCHITECTURE



5. RESULTS

SYSTEM OUTPUT

- ✅ 3D Kidney/Nephron visualization
- ✅ Smooth Meta Quest 2 interactions
- ✅ Real-time function/disease simulation

VR INTERFACE ACTIVE

VALIDATION

- ✅ Improved anatomical retention
- ✅ User-friendly VR environment
- ✅ Responsive learning experience

VR VALIDATED

Success Outcome